BrightLearn

Data Analytics

CASE STUDY

Car Sales Analysis

Bright Motors

Business Insights for a New Head of Sales

Role: Junior Data Analyst | Dataset: Bright Car Sales

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1. INTRODUCTION & INSTRUCTIONS

1.1 INTRODUCTION

You have been provided with a dataset titled "Bright Car Sales", which captures daily transactional and pricing information for vehicles sold by Bright Motors.

Bright Motors has recently appointed a new Head of Sales, whose mission is to expand the dealership network, improve sales performance, and optimize inventory. Your role, as a Junior Data Analyst, is to extract actionable insights from the historical car sales data and prepare a presentation that will help guide future sales and marketing strategies.

2. OBJECTIVE

Use your data analytics, SQL, and data visualization skills to help Bright Motors understand:

- Which car makes and models generate the most revenue
- The relationship between price, mileage, and year of manufacture
- Which regions or locations have the highest sales volumes
- Emerging trends in customer purchasing preferences
- Recommendations to increase dealership profitability and efficiency

3. TOOLS ALLOWED

You may use any of the following tools (or equivalents):

Coding / SQL	Snowflake, Microsoft SQL Server, Databricks, Google BigQuery, MySQL Workbench
Visualization	Microsoft Excel, Power BI, Google Sheets, Google Looker Studio
Presentation	Microsoft PowerPoint, Canva
Project Planning	Miro, Figma

4. TASKS (Detailed)

Task 1 Planning & Architecture (Miro or equivalent)

Create a Miro board (or any whiteboard tool) that outlines:

1. **Data Flow / Architecture Diagram showing:**
    - Source: car sales dataset (CSV/Excel file)
    - ETL Pipeline: data cleaning (removing duplicates, handling missing values, currency formatting)
    - Storage: Snowflake or chosen SQL database
    - Analysis Layer: SQL and visualization tools (Excel / Power BI)
    - Presentation Layer: PowerPoint / Canva
2. **Key Insights to Deliver:**
    - Revenue by car make and model

- Sales distribution by year and fuel type
- Regional performance (city or province)
- Average selling price trends over time

3. Key Calculations:

Total_Revenue	Selling_Price × Units_Sold
Profit_Margin	(Selling_Price − Cost_Price) ÷ Selling_Price × 100
Grouping By	Make, Model, Year, Region, Fuel_Type

Task 2 Data Processing in Snowflake (or chosen SQL platform)

Steps:

- Convert the provided Excel file into a CSV file.
- Load the CSV file into Snowflake (or equivalent).
- Perform data cleaning and transformations:
  - Convert text-based prices (e.g., '15,000') to numeric format.
  - Create a new column: total_revenue = selling_price * units_sold.
  - Calculate profit_margin and categorise cars by performance tiers (High Margin, Medium Margin, Low Margin).
  - Group transactions by time periods (month, quarter, or year).

Task 3 Data Analysis & Visualization (Excel / Power BI / Google Looker Studio)

- After transforming your data in Snowflake, export the processed dataset to Excel or connect Power BI directly to your database.
- Use slicers for region, fuel type, and year to allow interactivity.

Task 4 Presentation to the Head of Sales

- Create a professional presentation that summarises your findings and recommendations.

5. SUBMISSION GUIDELINES

Required files:

1	**Architecture Diagram**	Miro Plan / Architecture Diagram (image or link)
2	**Processed Dataset**	car_sales_processed.xlsx
3	**Presentation File**	BrightMotors_Presentation.pdf
4	**SQL Script**	car_sales_queries.sql

6. TIPS & NOTES

- Ensure all prices are converted to numeric formats (remove commas or currency symbols).
- Handle missing or inconsistent values carefully.
- Use clear and consistent column names in Snowflake.

- Include short insight notes in your dashboards.
- Your presentation should be visually appealing and concise.

End of Case Study

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